Signed paired-resultant trace and a P-192 degree audit

Signed paired coordinates -> quadratic block
Keep x(jG), x(jQ), and y(jG)*y(jQ)

Baby algebra -> resultant norm and trace coefficient
h(T), A(T), B(T); denominator inversion modulo h

Degree-dependent contraction -> explicit cost
Charge encoding, polynomial degree, point inputs and pole precision

P-192 audit -> bounded encoding exclusions
Known small-scalar controls pass; three declared target/CM fixtures fail caps8/16

Construction, 2026-11-08. This round constructs a signed baby/giant algebra adapter for the cyclic marked trace from Round56. It keeps both paired coordinates and their relative sign, contracts each block by a quotient-algebra trace, and checks its norm interpretation through determinants. An actual P-192 rank audit then tests the low-degree rational encodings needed for a cheaper contraction. All claims are scoped derivations or exact computations, pending trusted review. [pcc:unverified]

Signed coordinates. For E:y^2=f(x)=x^3+ax+b, use the product-Kummer coordinates kappa(A,B)=(x(A),x(B),y(A)y(B)), with z^2=f(x1)f(x2). They identify simultaneous negation and retain the relative sign between factors. Given S=(S1,S2), T=(T1,T2), write si=x(Si), ti=x(Ti), di=(si-ti)^2, Hi=(f(si)+f(ti))/di-si-ti and Ki=2*y(Si)*y(Ti)/di. Assume both di nonzero. For w(A,B)=x(A)x(B), the two roots w(S+T),w(S-T) have trace sigma=2*(H1*H2+K1*K2) and product tau=(H1^2-K1^2)*(H2^2-K2^2). Here K1*K2=4*z(S)*z(T)/(d1*d2). Expanding the two products proves the quadratic Z^2-sigma*Z+tau. [pcc:unverified]

Why relative sign matters. Keeping the x-coordinates and reversing only the second y-coordinate of T changes sigma by16*z(S)*z(T)/(d1*d2). It instead pairs S1+T1 with S2-T2, and S1-T1 with S2+T2. Thus two independently signed x-lines do not specify the required mixed trace. The native controls check both the intended and crossed pairings, including sixteen full-width P-192 pairs. [pcc:unverified]

Index construction. For r>=5, let m=floor(sqrt(r-1)/2), n=floor((r-1)/(4m)), I={2m(2i+1):0<=i<n}, J={2j+1:0<=j<m}, and K the remaining odd indices above4mn and below r. The signed sets +/-(I+J), +/-(I-J), +/-K and0 partition Z/rZ. For each giant i and sign, take S=(P+/-iG,R0+/-iQ); babies are T_j=(jG,jQ). The pair quadratics cover the block, and the identity/tail terms complete C_Gamma(P,R0). All declared probes avoid the pair denominators. This implements generic trace evaluation; substitution at a pole requires cancellation or controlled Laurent precision. [pcc:unverified]

Baby algebra and resultant. Put X_j=x(jG), A(X_j)=x(jQ), B(X_j)=y(jG)y(jQ), and h(T)=product_j(T-X_j). Lagrange interpolation gives A,B of degree below m, and the block coefficients sigma,tau lie in L=F_(p^2)[T]/h after denominator inversion. The norm polynomial is Res_T(h(T), Z^2-sigma(T)Z+tau(T)), monic of degree2m. Its next coefficient is minus Tr_L(sigma). A coefficient-only contraction gives Tr_L(sigma)=[T^(m-1)](sigma*h' mod h), so the trace route need not expand the norm polynomial. The native checker compares this coefficient with a multiplication-matrix trace, the individual pair sums and the direct cyclic trace; three field specializations of each norm match multiplication-matrix determinants. [pcc:unverified]

Cost and the interpolation dependency. With generic degree-(m-1) A,B, each giant requires arithmetic and one inversion modulo h, costing O(M(m)log m) with fast polynomial arithmetic. There are2n giant blocks, giving O(n*M(m)log m), plus interpolation, point construction and tails. For m,n of order sqrt(r), this is quasi-linear in r as a field-operation bound. The native polynomial backend is schoolbook; its operation count is not that of a fast backend. Direct pair checks, norm expansion and complete cyclic enumeration are verification work, additional to the proposed trace route. [pcc:unverified]

Conditional compression. If A=N_A/D_A and B=N_B/D_B have all numerator/denominator degrees at most delta, with denominators nonzero on the baby nodes, clearing the block trace denominator gives numerator and denominator degrees at most2+3delta. A balanced rational sum over the giants therefore costs O(M(n*(delta+1))*log n), followed by one reduction/inversion modulo h and the trace coefficient. Include the cost of obtaining that encoding and all O(m+n) explicit point inputs. Constant delta would reduce this field contraction to about square-root size, while the explicit point-input term remains. Neither low degree nor implicit batch construction is assumed from the class number. [pcc:unverified]

P-192 audit. Using the full-width p,r,a,b,G and public Q from Round52, freeze the first48 positive odd indices. For degree cap8, use24 training nodes and8 heldout nodes; for cap16, use40 and8. For each of A and B, solve N(X_j)-value_j*D(X_j)=0 with both degrees capped. The rectangular matrices have18 or34 columns. A nonzero square minor proves full column rank and excludes every nonzero rational pair at that cap, even on the training nodes. Conversely, every retained fit must have nonzero denominators and pass all heldout points. The known scalar controls additionally match their complete multiplication-map rational functions by polynomial cross multiplication. [pcc:unverified]

Audit outcomes for both paired functions. G,-G,2G fit both caps. 3G is excluded at cap8 and fits cap16. The public target Q, the Round25 order-five mu*G, and the Round25 principal-relation lambda*G are excluded at both caps, for each coordinate. All28 systems

are retained:14 positive rational identities and14 full-column-rank witnesses. A separate native verifier recomputes384 points with Jacobian arithmetic and checks every nonzero minor using matrix powers and Newton identities rather than row elimination. These are exact bounded-encoding exclusions, not a statement that those targets have no other representations. [pcc:unverified]

P-192 bounded rational encoding audit (both paired functions)

	Degree cap8:18 columns	Degree cap16:34 columns
G	fit: ranks 10/18, 12/18	fit: ranks 18/34, 20/34
-G	fit: ranks 10/18, 12/18	fit: ranks 18/34, 20/34
2G	fit: ranks 13/18, 15/18	fit: ranks 21/34, 23/34
3G	excluded: ranks 18/18, 18/18	fit: ranks 26/34, 32/34
public_Q	excluded: ranks 18/18, 18/18	excluded: ranks 34/34, 34/34
mu_G	excluded: ranks 18/18, 18/18	excluded: ranks 34/34, 34/34
relation_lambda_G	excluded: ranks 18/18, 18/18	excluded: ranks 34/34, 34/34

Full rank excludes the capped rational pair. Positive fits also pass heldout/global-map checks.

Native panel. Over F11,r7 and F17,r19 use the complete preceding target/probe panels. Add E/F211:y^2=x^3+7, G=(3,33), with directly counted prime order199, twelve frozen targets and two probes. Across the96 traces, the native checks cover2820 pair quadratics,2820 reversed-sign rejections,588 trace coefficient blocks and1764 determinant/norm comparisons. The actual P-192 extension-field panel checks sixteen more paired quadratics and their crossed-sign alternatives, using P=G+S,R=6S and the saved quadratic shift. No full P-192 trace or pole evaluation is included. [pcc:unverified]

Implication for the CM mechanism. The costed relation scalar and its order-five projection do not supply the tested cap8 or cap16 single-chart encodings. This audit keeps the curve, CM field and subgroup fixed; it does not estimate an L-value, compare class numbers or test a seed-selection law. The next construction is a joint-coordinate contraction or multi-chart encoding whose size is controlled by the horizontal relations, together with an implicit batch representation if the explicit m+n point term is to be reduced. Preserve the relative sign and the pole-precision obligation in that interface. [pcc:unverified]

Formal scope. lean/Pair.lean proves the integer polynomial trace identity using Lean4.19.0 and standard foundations propext and Quot.sound. It does not formalize the finite-field, Kummer, resultant or rank-certificate bindings. Those arguments remain written and natively checked. The source, failed parser invocation, corrected run, exact records, replay and mutation outcomes are retained. No runtime or speedup comparison is made. [pcc:unverified]

Sources checked. BFLS, https://arxiv.org/pdf/2003.10118 , Lemmas4.3,4.9 and sections4.13-4.16: single-coordinate products, jets, and the additional-coordinate issue for abelian varieties. Robert, Efficient algorithms for abelian varieties and their moduli spaces, https://www.math.u-bordeaux.fr/~damienrobert/pro/publications/academic/hdr.pdf , section4.3 pages68-69: traces via formal kernel points/resultants and the cost of representation/interpolation. These motivate the adapter; their bounds are not assigned to its paired input without accounting for A,B. Sources checked (KB): MCP tools unavailable and migrated wrapper absent. No novelty or universal absence claim follows from this search. [pcc:unverified]